Packet Tracer

IoT Devices

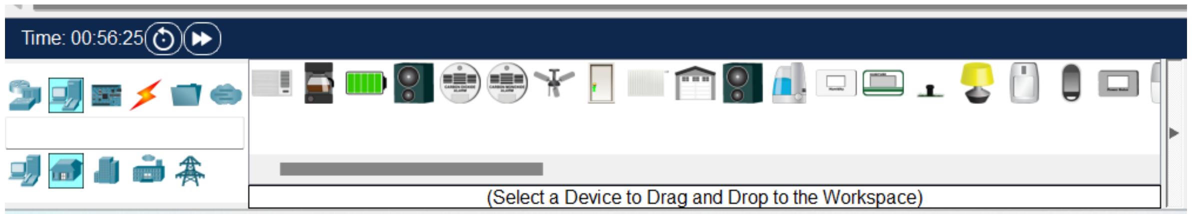
Config and Programming

Agenda

- IoT Devices in Packet Tracer
- Programming IoT Devices in PT
- Linking PT simulation to real world!

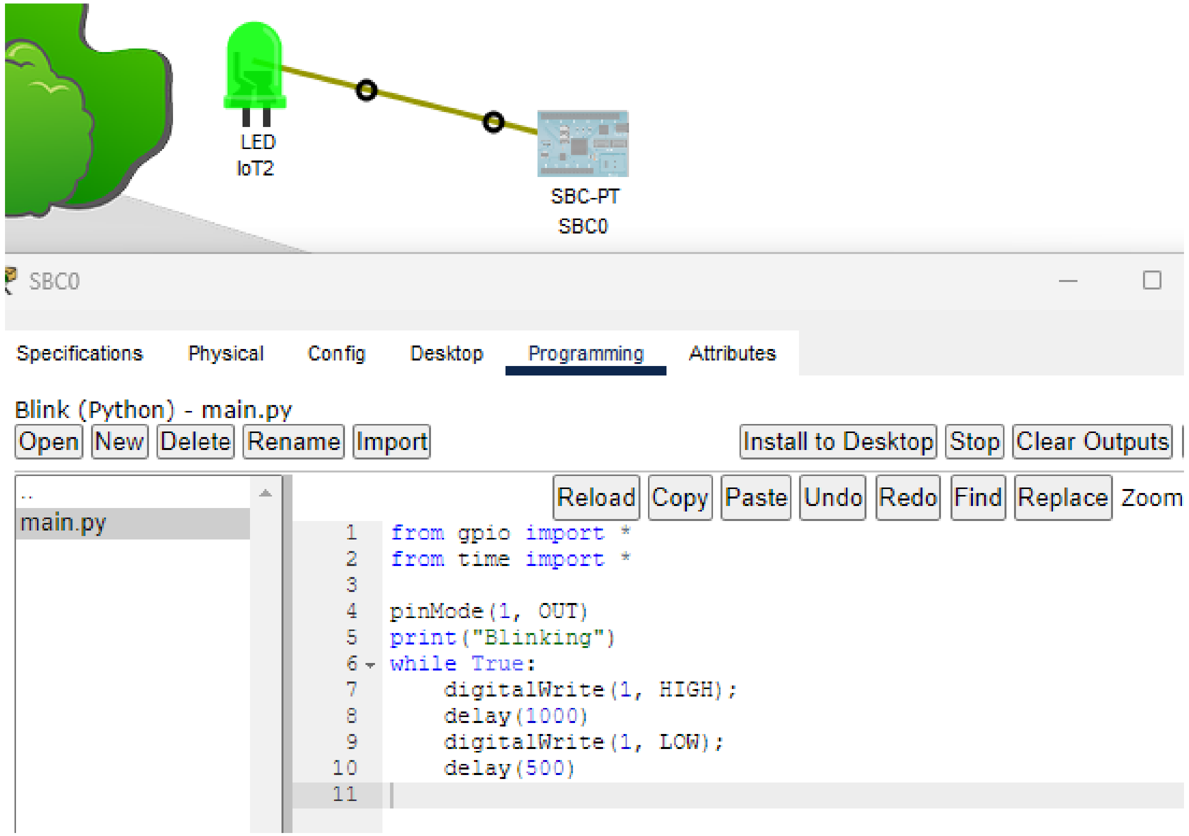
IoT Components in Packet Tracer

- Packet Tracer can be used as a prototyping tool for IoT.
- Packet Tracer can simulate smart networks that make use of IoT devices.
- Many IoT devices for a Smart Home/Industrial/Power Grid network.
- Can build your own "custom" IoT device
- Direct interaction with a device:
    - Hold down the Alt key and click on the device to turn it on or off.
    - Control remotely over the network via a web browser



MCU/SBC/Sensor and Actuators

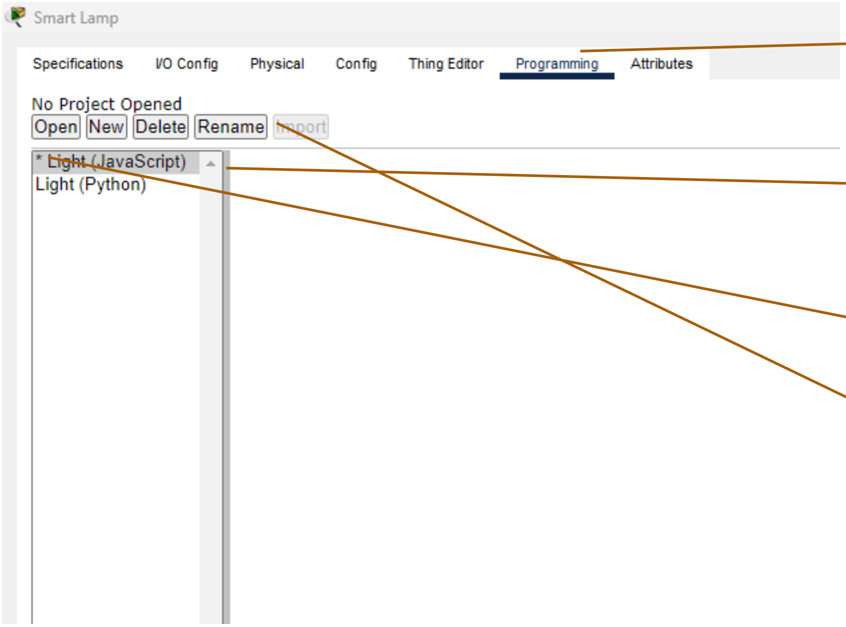
- PT contains a set of common sensors and actuators (components)
  - LED, Button
- Components can't be directly connected to the Network
- Connect to Single Board Computer or Microcontroller, you can create/prototype electronic circuits.
- Connect them using IoT Custom Cable



Program a Device or Custom Component

- Packet Tracer provides support for JavaScript, Phyton and Visual Programming language called Blocky for most IoT devices/components, including Custom devices
- Access the Programming environment via advanced->Programming tab
- Usually, a default example Project(s) present
- Can create new programming project or edit an existing one

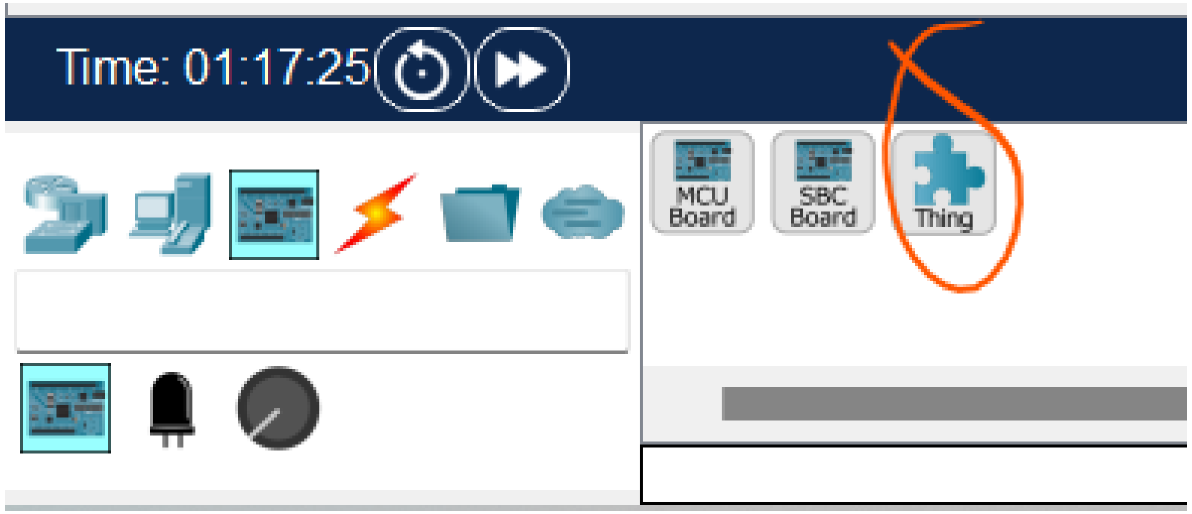
Programming: Projects



1. Select Programming Tab
2. Left Panel shows projects available on device
3. Asterix indicates which Project is running
4. Use buttons to Open/Delete/Create/Stop/Start projects

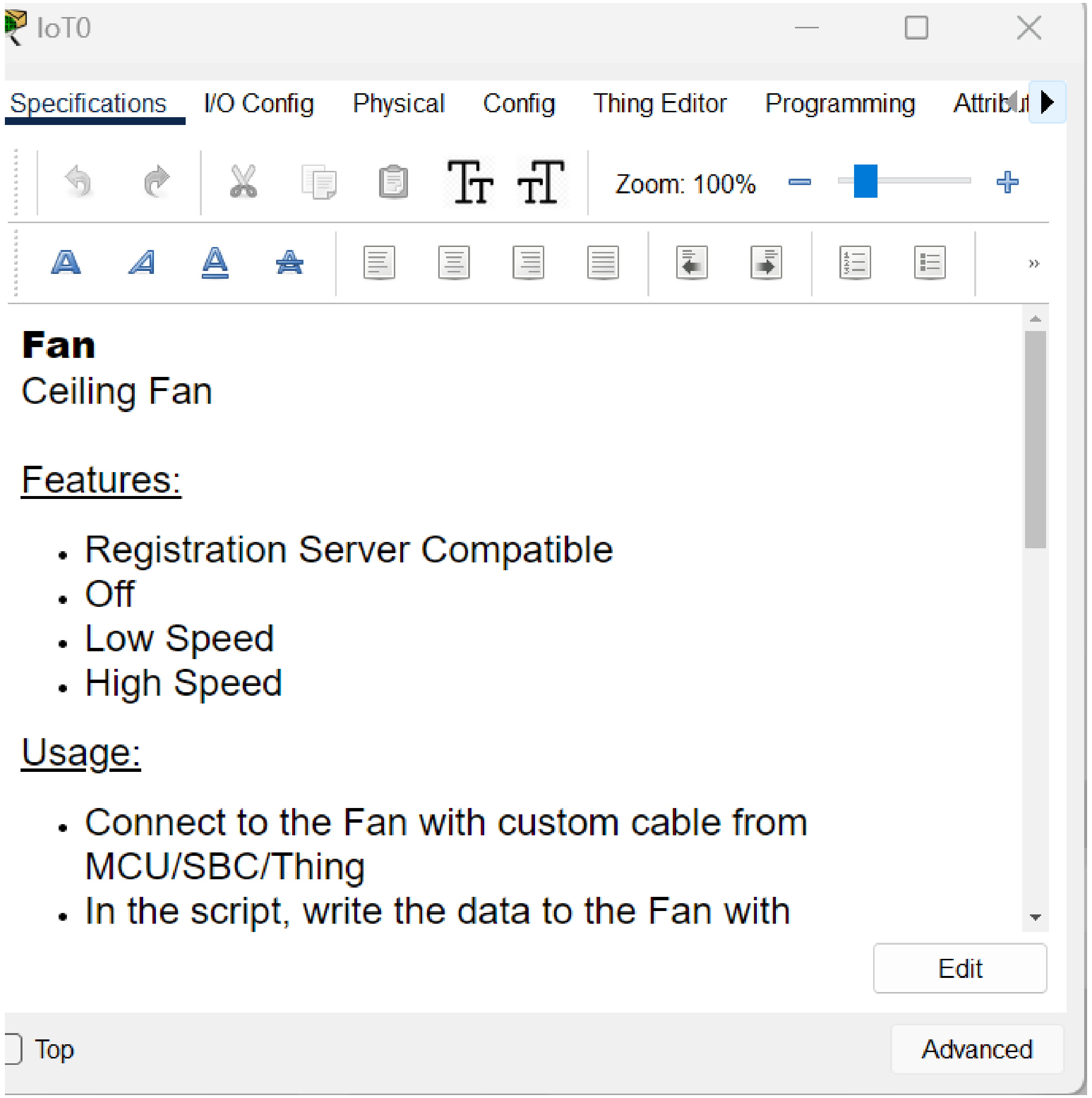
Create a new Component

- To Create a new Thing:
  - Add a Thing from Components list
  - Click the Advanced to access the /Thing Editor
  - Associate the new graphics to their respective states(e.g. on/off)
  - Click on the Config tab to select the network adaptor to be used to connect to the network (if applicable).
- Save it for later:
  - Select Tools/Custom Device Dialog from the toolbar.
  - Click on Select and click on the Thing to be saved.
  - Modify the template and description.
  - Click on the type of new Thing.
  - Click on Add - The new template will be saved in the PT template file on the local disk and the customized Thing will now display with the other sensors.



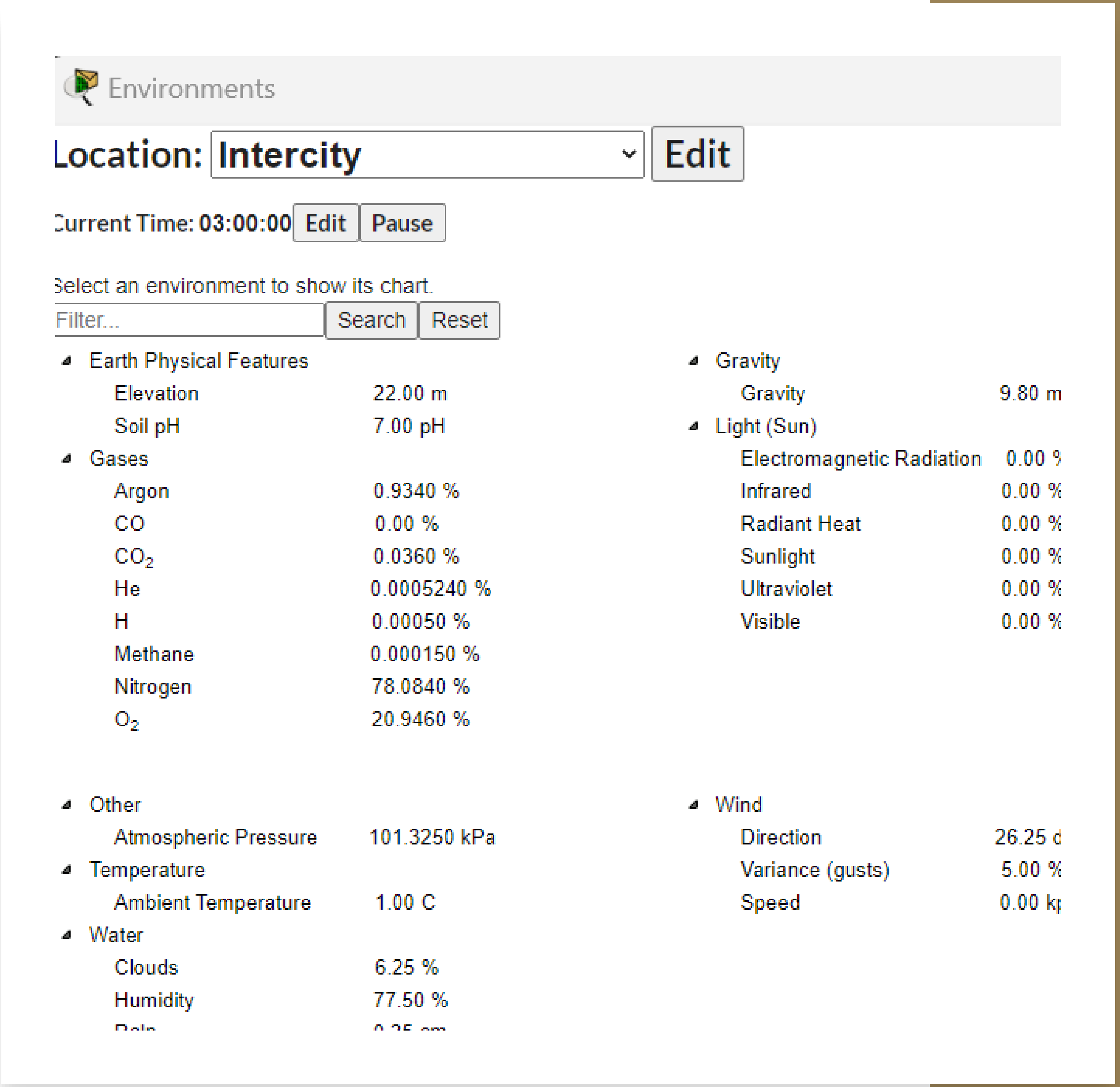
IoT Components: Configuration

- The Config window of a device allows you to:
    - Specifications - describes the features, usage, local and remote control of the device.
    - Physical - available modules and power connections.
    - Config - show display name, serial number, network configuration and IoT server.
    - Attributes - display the devices attributes such as MTBF, power consumption and cost.
- Advanced config also possible including programming/behaviour



Packet Tracer Environment Controls

- The Physical Workspace uses containers.
- Each container (intercity, buildings, and wiring closets) can have their own set of environmental values.
- There are 24 default environmental elements, such as temperature, rain, water level, wind speed, and snow.
    - A heater will increase ambient temperature when turned on.
    - A smoke detector can be used to trigger an alarm when the smoke in environment increases to a certain point.



Programming: Creating a new Project

- Select the Template to suit
  - Many available in Python/Javascript
- Modify the code to suit your means
  - Good idea to look at similar devices code as an example